



Community Planning & Permitting

Courthouse Annex • 2045 13th Street • Boulder, Colorado 80302 • Tel: 303.441.3930 • Fax: 303.441.4856
Mailing Address: P.O. Box 471 • Boulder, Colorado 80306 • www.bouldercounty.org

BOULDER COUNTY PLANNING COMMISSION PUBLIC HEARING

August 17, 2022 at 1:30 p.m.

Hearing to be Held Virtually due to COVID-19

STAFF RECOMMENDATION

STAFF PLANNER: Pete L'Orange, Planner II

Docket SU-22-0003: Dowe Flats Mining and Reclamation Extension

Proposal: Special Use/Site Specific Development Plan review to amend an existing Special Use approval (SU-93-14) for limestone/shale open mining/quarrying located at the Dowe Flats quarry as follows: extend approved mining activities for an additional 15 years; reduce permit area from 1,911 acres to 709 acres; conclude cement plant operations at the facility located south of Highway 66 within the same 15-year timeframe; conduct concurrent reclamation of wildlife habitat.

Location: 13301 55th Street, Parcel #120316000050, located approximately 0.5 mile north of the intersection of N. 53rd Street and state Highway 66, in Sections 9, 10, 15, and 16, Township 3N, Range 70W.

Zoning: Agricultural (A) Zoning District

Owner: CEMEX Inc.

Applicants: Pamela Franch Hora (Primary); Tetra Tech
John Heffernan; CEMEX, Inc.
Therese Glowacki; Boulder County Parks & Open Space

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SUMMARY

The applicants request Special Review and Site-Specific Development Plan review to amend an existing Special Use approval (SU-93-14) to permit an existing limestone/shale open mining and quarrying operation located at the Dowe Flats quarry. The original SU/SSDP approval allowed for the open mining activities to occur for a period of 25 years, starting once mining activities had commenced (which was defined as the movement of at least 100,000 tons of mined material). That 25-year period is scheduled to end in 2022. The original approval also included a number of conditions on, and commitments from, the applicant to mitigate and offset the impacts of the mining activities. The applicant has now requested that the SU/SSDP approval be extended for an additional 15 years. The applicant has proposed additional commitments of record in order to mitigate the land use impacts of the proposal as part of this application.

With the recommended conditions, staff finds the request can meet the Special Review Criteria in Article 4-601 of the Boulder County Land Use Code (the Code) and recommends the Planning Commission recommend conditional approval to the Board of County Commissioners.

HISTORY

Dowe Flats Quarry

The applicants have requested an extension of the mining activities at the Dowe Flats quarry, located north of State Highway 66, for an additional 15 years (see Figure 1 below).

In 1994, the Board of County Commissioners approved a Special Use/Site-Specific Development Plan application for an open mining proposal at the Dowe Flats quarry (SU-93-14). This approval allowed the applicant to conduct mining activities on site to extract approximately 760,000 tons of material a year, with a maximum 25-year mining period. The mining activities at Dowe Flats are also approved through a Colorado State mining permit (M1993-041).

SU-93-14 approved mining activities to progress from the southern portion of the project area to the north, with reclamation occurring as mining activities were completed in each section. Mined materials were approved to be transported from the Dowe Flats quarry across Highway 66 to the cement plant at the Lyons Quarry site via an enclosed conveyer system; transportation of materials was temporarily allowed by truck until the conveyer system was built.

As part of the SU/SSDP approval, the applicants were required to comply with 16 conditions (see Attachment D for the full list of conditions). As one of those, the applicants were required to submit, for review and approval by County Staff, plans for the management and restoration of the land impacted by the mining activities. This was also a requirement of the State of Colorado Division of Reclamation, Mining and Safety permit.

Additionally, the applicants were subject to periodic reviews of the mining operations to ensure that the conditions of approval were being met. These reviews found the operations generally in compliance with the terms and conditions of the approval (See full results in Attachment E). Other conditions of approval included the establishment of non-development covenants on the project area, the donation by the applicant of several conservation easements, the construction of the enclosed conveyer system over Highway 66, and the relocation of some utilities and public rights-of-way.

Currently, the Dowe Flats quarry continues to operate under the existing Boulder County SU approval and State of Colorado permits. To date, approximately 158 acres of the project area have been reclaimed.

Lyons Quarry and Cement Plant

Separate from the mining activities at the Dowe Flats quarry, CEMEX also conducts mining and cement processing activities south of Highway 66. This southern quarry is known as the Lyons

Quarry (see Figure 1 below). The following information on the Lyons Quarry and the associated cement plant is to provide context on the cement processing activities. The operations at the Lyons Quarry and cement plant were not approved under SU-93-14 and are **not** part of the current application to extend mining activities at the Dowe Flats quarry site. The plant and activities on the southern parcels operate under the Land Use Code and state law as nonconforming uses. As nonconforming uses, the owners or any subsequent owners have a right to continue operating the plant at this time.

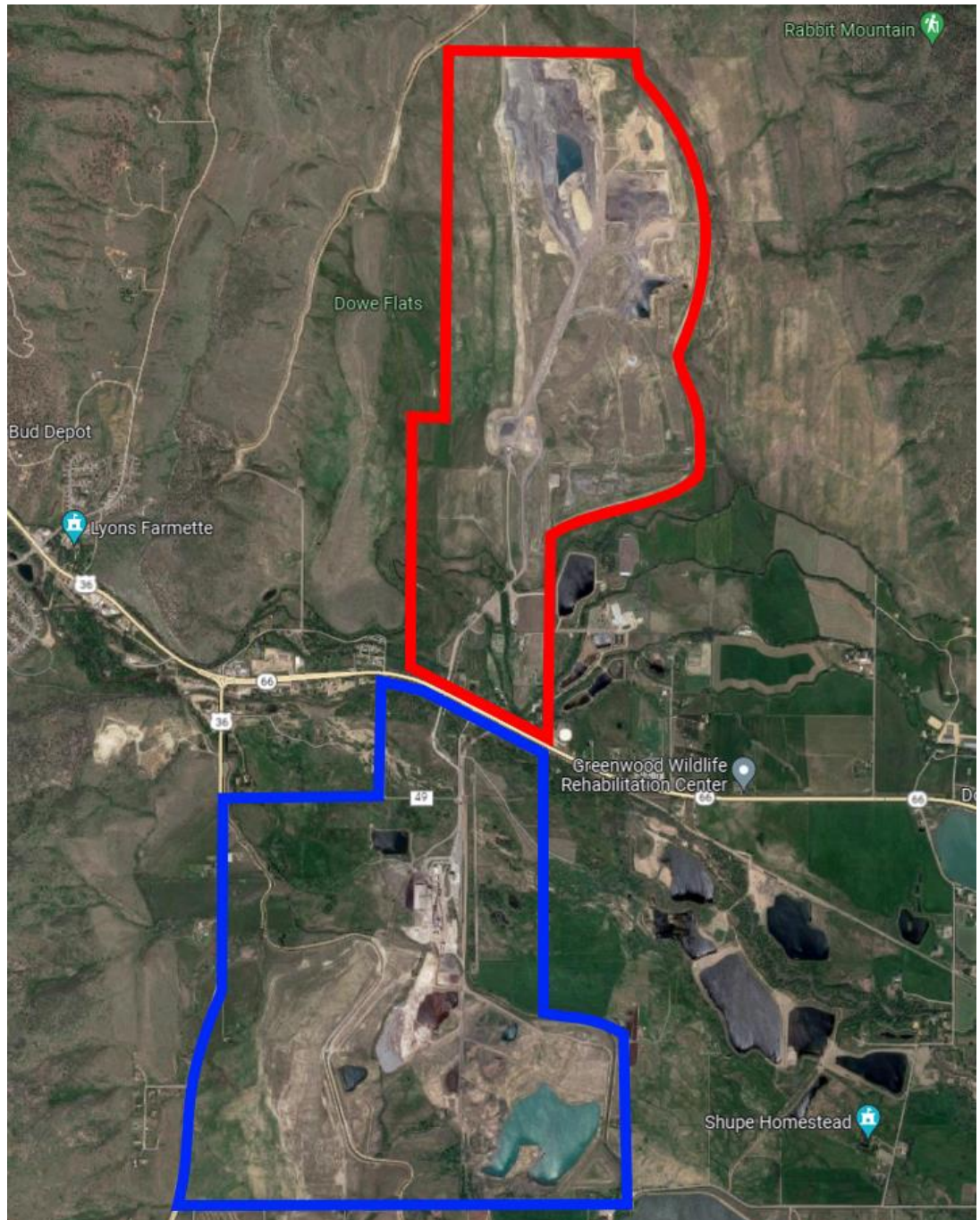


Figure 1: Dowe Flats quarry shown in red; Lyons Quarry shown in blue.

DISCUSSION

The subject property, the Dowe Flats quarry, is located north of State Highway 66, between the City of Longmont and the Town of Lyons. The property is in the Agricultural zoning district. The Dowe Flats quarry open mining use area includes a total of 15 parcels (Parcels 120316000050, 120316000046, 120316001001, 120316001002, 120316001003, 120316001004, 120316001005, 120316001006, 120316001007, 120316001008, 120316001009, 120316001010, 120316001011, 120316001012, and 120316001013). The open mining pits are located on northern-most parcel, 120316000050, while the other parcels are primarily used for the on-site processing of materials and transporting the materials via the conveyor system. The total area encompassed by the proposed extension request is 709 acres.

The subject property has the following designations as determined by the Boulder County Comprehensive Plan: Adjacent to County Open Space on the west, north, and east; Environmental Conservation Area – Rabbit Mountain; High Biodiversity Area – Rabbit Mountain (ranked “B1,” of outstanding significance; limited on parcel); Natural Area – Rabbit Mountain (limited on parcel); View Protection Corridor; Nearby Agricultural Ditches – Supply, Highland, Rough & Ready, Palmerton; Nearby Critical Wildlife and Preble’s Meadow Jumping Mouse habitats – on St. Vrain Creek, and ditches.

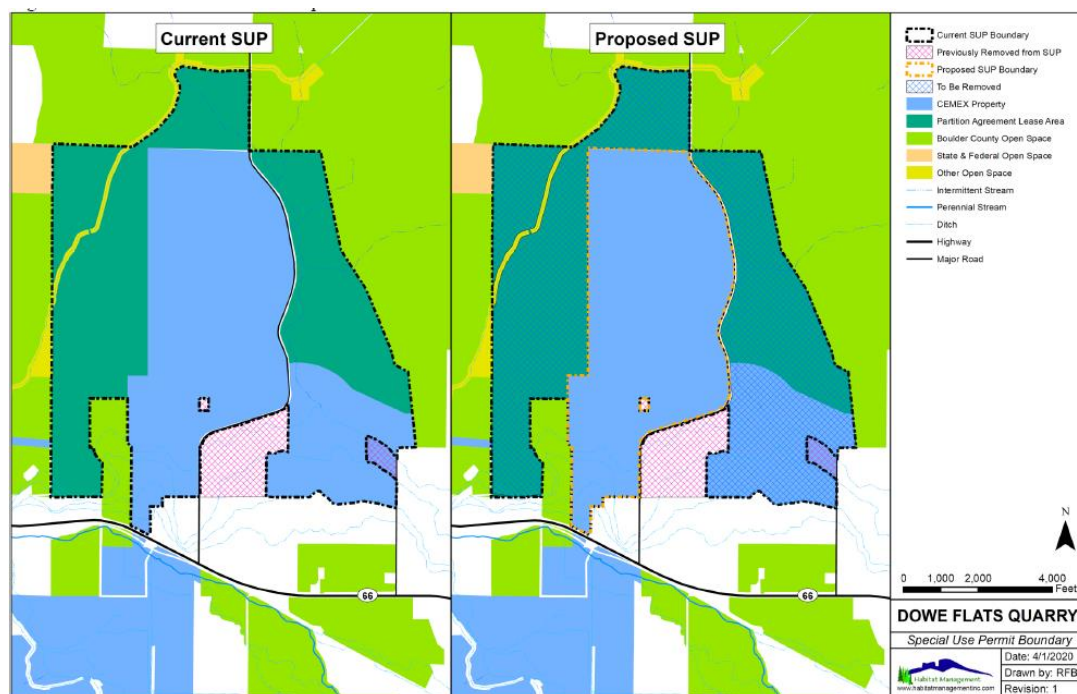


Figure 2: Current and proposed boundaries as provided by the Applicant. Note: “SUP” stands for “Special Use Permit”, usually referred to as SU/SSDP in Boulder County.

The originally approved SU/SSDP boundary encompassed an area of 1,911 acres. Approximately 101 acres have been removed from the boundary over the years due to property sales. The current SU/SSDP area is approximately 1,810 acres. The applicant is proposing, as part of the requested extension to further reduce the boundary to an area of 709 acres. The areas west, north, and east of the proposed new boundary were included in the original approval as buffer areas, much of which is no longer necessary for the current mining activities or the proposed extension of those activities. In addition, much of that land has been transferred to the County. The applicant submitted a map

showing the current SU/SSDP boundary and the proposed boundary (see Figure 2 above).

The current Boulder County approval for mining at Dowe Flats (SU-93-14) included a specific timeframe for the mining activities, which is to end in 2022. The applicant has now applied for Special Use/Site Specific Development Plan review to amend the existing Special Use approval (SU-93-14) to permit the Open Mining use to continue at the Dowe Flats quarry for an additional 15 years.

During this time, the applicant would continue to extract resources from the existing mining pits on the property. The footprint of the mining pits on the subject property would not increase. Rather, the applicant would simply mine deeper in the same location. The applicant has offered the following commitments of record as part of the proposal, as stated in the submitted application materials (see Attachment A):

- A reduction of the purchase prices for Boulder County's existing options to acquire real property north of Hwy. 66 to zero dollars;
- ~~An additional option to Boulder County to purchase (when mining at Dowe Flats is completed) four additional parcels totaling approximately 200 acres around the perimeter of the Dowe Flats mine, at a purchase price of zero dollars;~~
- ~~An increase in the required rental payments by CEMEX to Boulder County pursuant to the existing buffer lease for properties around the Dowe Flats mine from the current amount of \$1,000/year to an increased amount of \$400,000/year;~~
- ~~The dedication of a permanent, non-exclusive recreational trail easement to Boulder County for the benefit of Boulder County Parks & Open Space along the south bank of the St. Vrain River or another mutually agreed location;~~
- ~~A commitment by CEMEX to permanently conclude ongoing cement plant operations at its facility south of Hwy. 66 within the same 15-year timeframe for completion of mining operations (plus reclamation) at Dowe Flats; and~~
- ~~An additional option to Boulder County to purchase up to approximately 830 acres of CEMEX property surrounding CEMEX's cement plant south of Hwy. 66 at a current price of \$17,000 per acre with a 2.0% annual escalator, upon condition that the Town of Lyons provide consent to Boulder County's acquisition of any lands within the CEMEX Municipal Facilities Area per the 2012 CEMEX Area IGA Map, and provided that CEMEX would reserve an access and utility corridor to/from Hwy. 66 for the benefit of its retained properties.~~

The applicant would continue to be subject to the conditions of approval as set forth in SU-93-14, as well as all applicable conditions and approvals required through the various required State permits. These include the requirements to carry out the required reclamation on the subject parcel once the mining activities have ceased.

The reclamation plan as approved through SU-93-14, requires the operator to restore the site in a way that is compatible with the surrounding area. Per the application materials (see Attachment A) submitted:

"The currently approved reclamation plan includes two basic vegetation communities: grassland and wetland. The grassland is a somewhat diverse mixture of native range grasses with a small component of wildflowers. The 20-acre wetland, as planned, is to be 98% cattail and 2% rushes. In 1993, when this reclamation plan was approved, at the request of the County, the reclamation plan was geared towards the development of habitat for a single primary species, prairie dogs, which in turn provide food for raptors. While it is important to think about providing some prairie dog habitat, there is already a substantial amount of prairie dog habitat in the area. This site has the potential to provide habitat for many additional species. Because of the location of the property in relation to other open space

land, the site could effectively function as a corridor that expands and connects habitats existing on nearby properties.”

As part of the requested 15-year extension, the applicant has committed to working with Boulder County to revise and improve the reclamation plan to better incorporate the original topographical and vegetative diversity of site in order to provide habitat for a greater variety of local fauna. In addition, sections of the site could be planted with pollinator-friendly vegetation to enhance insect populations. The applicant has also proposed including a diverse wetland habitat with a variety of sedges, rush, forbs, and shrubs interspersed with open water that could be incorporated into the reclamation plan.

As detailed in the criteria review below, staff finds that the proposed extension of the existing Special Use/Site Specific Development Plan (SU-93-14) for a period of 15 years can meet the Special Review Criteria in Article 4-601 of the Code, with the recommended conditions of approval.

REFERRALS

This application was referred to the typical agencies, departments, and adjacent property owners. The original deadline for agencies and departments to respond was June 10, 2022; at the request of the Town of Lyons, that deadline was extended to July 22, 2022.

County Development Review Team – Access & Engineering: This team reviewed the proposal and responded that legal access to the subject property had been demonstrated via State Highway 66 (SH 66), also known as Ute Highway, a Colorado Department of Transportation (CDOT) owned and maintained right-of-way (ROW). They concurred with the trip generation estimates provided in the Pre-Application Methodology Statements (PAMS) dated October 17, 2021 of approximately 20 Average Daily Trips (ADT) being generated by the continued mining and reclamation use on the subject property. Staff does not anticipate the mining and reclamation use will have negative impacts to the surrounding transportation network. Staff has no concerns with the continued use of the subject property for mining and reclamation activities.

~~**County Public Health Department:** This department reviewed the proposal and responded that it would be willing to support the mining extension at Dowe Flats for an additional 15 years if the applicant can commit to completing the upgrades and improvements to the plant necessary to address long standing and ongoing fugitive dust and maintenance issues. BCPH requests Cemex completes the Lyons plant upgrades. Management, along with BCPH staff, has identified the following as necessary future improvements: conveyor belt enclosures, solid closures where there are curtains, Clinker Pit improvement enclosure with direct conveyance to the A-Frame storage (this is one of the biggest problem areas), broken baghouse detectors, paving the east haul road and improved camera surveillance of the plant.~~

~~**Boulder County Parks & Open Space:** This department reviewed the application materials and determined the proposed mining at Dowe Flats will not extend outside the current disturbance area, rather it will result in mining deeper in the existing footprint. Therefore, the additional disturbance to environmental, cultural, and open space resources is limited. However, because the 15-year extension of mining activity does lengthen the 30-year life of the mine to 45 years, it prolongs the disturbance to the site and impacts to nearby lands. The department also determined that the commitments of record that CEMEX has proposed have important benefits. These are detailed in the May 2, 2022 letter to Dale Case and include the conclusion of plant operations on the same 15-year timeframe of the mine operation, the dedication of a non-exclusive trail easement for the St. Vrain Greenway Trail through the property, the option to purchase additional lands in the future around both the mine and plant sites, and finally monetary savings and income through modification of existing lease and option agreements.~~

~~BCPOS included detailed comments about the application as well as recommended conditions of approval in its referral (see the full referral in Attachment B for the specific recommended conditions of approval).~~

Boulder County Parks & Open Space – Natural Resource Planner: The Natural Resource Planner reviewed the application materials and determined the subject property has the following designations as determined in the Boulder County Comprehensive Plan: Adjacent to County Open Space on the west, north, and east; Environmental Conservation Area – Rabbit Mountain; High Biodiversity Area – Rabbit Mountain (ranked “B1,” of outstanding significance; limited on parcel); Natural Area – Rabbit Mountain (limited on parcel); View Protection Corridor; Nearby Agricultural Ditches – Supply, Highland, Rough & Ready, Palmerton; Nearby Critical Wildlife and Preble’s Meadow Jumping Mouse habitats – on St. Vrain Creek, and ditches.

In general, the Natural Resource Planner agrees with the application’s premise that impacts to natural resources would not significantly change, because: A) the Dowe Flats active mining footprint would not be modified; and B) future impacts can be interpreted as a continuation of existing, previously approved, conditions and impacts.

~~Additionally, if the proposal is approved with the “mitigating” commitments noted in the CEMEX letter from John Heffernan to Dale Case, dated May 2, 2022, long-term public benefits would significantly increase due to payment changes and the public disposition of many parcels after mining and plant closure. See the full referral comments in Attachment B.~~

Town of Lyons: The Town of Lyons firmly recommends that the Boulder County Planning Commission advise the Board of County Commissioners to deny the CEMEX mining permit extension application. The Town Board of Trustees, Staff, Boards and Commissions, and community concur.

The Town further asserts that it should have been a part of the discussions between Boulder County Open Space and CEMEX as they negotiated the terms of the real estate agreement, as the Town believes its Primary Planning Area as outlined in the 2012 IGA is directly affected. In addition, the entirety of CEMEX operations, both north and south of SH 66, fall directly within the Town’s state-mandated 3-Mile Planning Area.

The Town requests that the Planning Commission advise the Board of County Commissioners to deny the application and immediately reopen negotiations between CEMEX and the County, with the Town of Lyons as an active participant. The Town would further advise a short-term (six-month) mining extension be put in place.

The Towns states it believes there is a third option to consider, and it is one that every party can benefit from. The Town of Lyons states it would be amenable to a five-year mining and plant operation extension subject to specific conditions:

- Agreement to cease plant operations, decommission the plant and perform reclamation of the plant site and other agreed-to amenities by the end of the term;
- Renegotiation of IGA and ownership boundaries within the property;
- Agreement to annex and rezone the plant property from General Industrial to an agreed-upon zoning classification and discuss annexation and purchase of Lyons’ Primary Planning Area; and
- Assessment of the two reclamation bonds for Dowe Flats and the plant to ensure amounts are sufficient to cover current costs.

The Town does not believe that CEMEX can continue plant operations indefinitely if the SUP is extended. The current mining permit, M-1977-208, mandates that should mining operations cease

north of SH 66 at the Dowe Flats quarry, the cement plant south of SH 66 in turn must be demolished and the land reclaimed to “irrigated pasture”.

Adjacent Property Owners: Notices were sent to all property owners within a 1-mile radius of the subject property. All responses received as of August 9, 2022 are attached and summarized below. To date, staff received 63 public comments in response: 6 in support; 43 in opposition; and 10 comments classified as “other.”

The comments in support mainly focused on local jobs and the applicant’s commitment to close down the cement plant.

The vast majority of the comments received were in opposition. Most of these comments expressed concern about the environmental impacts of the cement plant and the impacts of the plant on the neighboring communities (mostly the Town of Lyons). The most common comments on the environmental impacts focused on carbon emissions and greenhouse gasses. Several comments mentioned past citations on the plant by environmental regulation agencies. Several questioned whether the cement plant would remain after operations at the Lyon Quarry ceased and expressed concern that the plant could simply be sold to another company who could continue to operate it. Likewise, there were a couple of comments questioning whether it would be economically viable for the cement plant to continue to operate if the Dowe Flats quarry was closed and suggesting that the closure of the Dowe Flats quarry would result in the cement plant closing as well, regardless of the applicants’ statement the plant could continue to operate “indefinitely.” Cited impacts to the neighboring communities included dust from the cement plant, light and noise pollution, traffic impacts, and negative impacts to local property values. Several comments also expressed concern about the impacts of the plant on the health of community members, most specifically respiratory issues.

The majority of the comments in the “other” category were members of the public asking that the Planning Commission hearing on this application for July 20, 2022 be delayed.¹ A few of the “other” comments were members of the public asking for clarification on technical items or for information on how to stay informed on the process.

Staff notes that many of the comments and concerns expressed by members of the public are specific to the cement plant located at the Lyons Quarry site. The Lyons Quarry and the cement plant are not part of the Docket SU-22-0003 application and are not currently under review.

Agencies that responded with no conflict: Larimer County Planning.

Agencies that did not submit a response: Boulder County Long Range Planning; Boulder County Building Official; Boulder County Assessor; Boulder County Attorney; Boulder County Conservation Easement Team; Northern Colorado Water Conservancy District; St. Vrain & Left Hand Water Conservancy District; Longmont Power and Communications; Poudre Valley REA; City of Longmont Planning & Development Services; Boulder Valley & Longmont Conservation District; Colorado Department of Natural Resources Division of Reclamation, Mining, and Safety; Colorado

¹ There was general misunderstanding by some members of the public that a hearing had been scheduled for July 20, 2022; no public hearing for this case was scheduled for that date. There was also a misconception that the original referral response deadline of June 10, 2022 for departments and agencies was also a deadline for comments from the public in general. When staff received comments of this nature, staff responded with a detailed explanation of the review process, how and when cases get scheduled to go before the Planning Commission and the Board of County Commissioners, and emphasized that the public comments would be accepted throughout the review process, including at the hearings by the Planning Commission and the Board of County Commissioners.

Department of Transportation; United States EPA Region 8; and Hygiene Valley Fire Protection District.

SPECIAL REVIEW CRITERIA

The Community Planning & Permitting staff has reviewed the proposal to amend an existing Special Use approval (SU-93-14) to allow limestone/shale open mining/quarrying activities at the Dowe Flats quarry for an additional 15 years pursuant to the Special Use Review standards in Article 4-601 of the Code, and finds the following:

- (1) *Complies with the minimum zoning requirements of the zoning district in which the use is to be established, and will also comply with all other applicable requirements;***

Open Mining uses are an allowed use through the Special Use Review process in the Agricultural Zoning District, where the subject property is located. The applicant proposes to amend the existing Special Review approval (SU-93-14) to extend the approved mining activities for an additional 15 years.

Article 4-508.C of the Boulder County Land Use Code defines Open Mining as “*The extraction of earth materials by mining directly from the exposed deposits or other materials. Exceptions to this use include those operations which fit the definition of limited impact open mining and excavations below finished grade for basements and footings of a building, retaining wall or other structures authorized by a valid building permit. The term open mining includes, but is not limited to, such processes as open cut mining, open pit mining, strip mining, quarrying and dredging.*” Additional provisions in the Code for Open Mining in 4-508.C.5, including:

- a. This use is not required to be located on a building lot, or comply with the minimum lot size requirement for the district in which it is located.*
- b. This use shall also be granted and maintain all applicable local, state, and federal permits.*
- c. Processing of the mined material (to the extent approved through the special use process) may occur on the parcel where the mining is situated, or on a parcel owned or leased by the mining parcel owner, lessee, or operator provided the parcel is located within 1,000 feet of the mining parcel.*

Staff finds that the current mining activities and the proposed extension of 15 years are consistent with the definition of Open Mining as described in the Code. The existing mining operations are considered to be “open pit mining,” and the proposed 15-year extension would not change that use.

Article 4-508.C.5.a states that Open Mining uses are not required to be located on a building lot or comply with the minimum lot size requirement for the zoning district in which it is located. The existing operations at Dowe Flats are on a site of over 35 acres, which is the minimum required lot size in the Agricultural zoning district. The proposed extension of operations will be located on the same site.

Article 4-508.C.5.b requires that an Open Mining Use be granted and maintain all applicable local, state, and federal permits. The applicants have obtained and maintained all applicable local, state, and federal permits to date. To ensure that the applicant and the mining operations continue to meet the conditions of Article 4-508.C.5.b, staff recommends that the Board include, as a conditional of approval, that the applicant obtain any new, or amend any existing, state, and federal permits as appropriate and necessary, and maintain any and all such permits for the duration of mining activities at the subject site.

Article 4-508.C.5.c allows for the processing of the mined materials to occur either on the same site as the mining activities or on another site, provided that other site is within 1,000 feet of the mining site. The materials currently mined at Dowe Flats are initially processed on the subject site, and then transported via the enclosed conveyor system to the facilities located at the Lyons Quarry site (as discussed above, the Lyons Quarry operations, including the cement plant and the processing of materials at the Lyons Quarry site, which are not part of this application and are not under review). The Lyons Quarry site is located across State Highway 66 from the Dowe Flats, a distance of approximately 150 feet. The extension of mining activities at Dowe Flats as proposed in this application would not result in any change to existing conditions related to the processing location. As such, this provision is met.

Therefore, as conditioned, staff finds this criterion can be met.

- (2) ***Will be compatible with the surrounding area. In determining compatibility, the Board should consider the location of structures and other improvements on the site; the size, height and massing of the structures; the number and arrangement of structures; the design of structures and other site features; the proposed removal or addition of vegetation; the extent of site disturbance, including, but not limited to, any grading and changes to natural topography; and the nature and intensity of the activities that will take place on the site. In determining the surrounding area, the Board should consider the unique location and environment of the proposed use; assess the relevant area that the use is expected to impact; and take note of important features in the area including, but not limited to, scenic vistas, historic townsites and rural communities, mountainous terrain, agricultural lands and activities, sensitive environmental areas, and the characteristics of nearby development and neighborhoods;***

The area surrounding the Dowe Flats quarry is primarily open space. There are a few residential and agricultural properties south of the quarry location clustered on several parcels located northwest of the corner of State Highway 66 and N. 53rd Street. There are also several municipal infrastructure facilities related to the City of Longmont's water treatment plant located on the east side of N. 53rd Street.

The application as submitted does not propose any changes to the existing use and does not propose new structures.

In considering the proposed removal or addition of vegetation, site disturbance, and any grading and changes to the natural topography that would result from this application, as required by this criterion, staff determined the utilization of the existing mining site is a unique situation, which may allow for this criterion to be met, subject to conditions as described below.

The Dowe Flats quarry operations since the original Special Use approval (SU-93-14) have already resulted in site disturbance, which is an expected consequence of the Open Mining use as approved. The continued mining activities which would result from the requested extension would be limited to the area in which site disturbance has already occurred. The application as submitted would not expand this existing "footprint." As such, staff finds that the proposed extension would not result in any significant increased impacts to the compatibility of the area from existing conditions; however, the requested extension would prolong the visual impacts which have already occurred. The original Special Use approval (SU-93-14) permitted the removal of the vegetation as it existed at that time, the site disturbance, and the grading and changes to the natural topography; this approval was conditioned (in part) on the eventual reclamation of the site post-mining.

This criterion also requires that the Board consider the potential impacts to scenic visits. The existing mining pit and most of the quarry operations are not visible from most public roads in the area. This is due both to the specific contours of the terrain surrounding the mining pits and the distance of the mining operations from public roads. The mining pit can be seen, however, from Dakota Ridge Road, north of the Dowe Flats quarry location. Likewise, the mining pit can be seen from the hiking trails at the Rabbit Mountain Open Space. Due to depth of the mining pit and distance from the mining pit to these public locations, however, most of the actual mining activities will not be visible to members of the public. The surrounding areas which have not been physically altered to date, would remain unaltered. As the Dowe Flats quarry site has already been impacted through the existing approval and the proposal would only allow for the mining activities to go deeper without expanding the footprint of the operations, continued mining operations at this specific site would not increase those impacts.

As for the impacts which have already occurred as a result of the mining activities as approved through SU-93-14, one of the conditions placed on that approval, and as required by various State Colorado permits, is that the applicant carryout approved reclamation efforts during and after mining operations at Dowe Flats. As addressed in the discussion above, the reclamation plans included in the application materials submitted are conceptual and are in need of refining to mitigate the impacts of the mining operations. Per the application materials (see Attachment A) and the applicant's subsequent reply to Boulder County Parks and Open Space's referral response (see Attachment C), the applicant agrees to work with Boulder County Parks and Open Space and the State of Colorado Division of Reclamation, Mining, and Safety to modify and refine the reclamation plans to ensure that the impacts resulting from the mining activities are appropriately mitigated post-mining operations at Dowe Flats.

~~In reviewing the application materials, Boulder County Parks and Open Space staff have stated they do not support the creation of wetlands and open water features in the reclamation plan for a variety of reasons. First, Parks and Open Space staff finds that returning the site as much as possible to its pre-mine landscape is the best for long-term ecosystem sustainability. The area was a grassland with associated wildlife, returning the site to that environment is best for the site and adjacent properties. Grassland reclamation of the site fits better into the landscape. Staff does not find that the 'varied' topographical reclamation proposed in the application will benefit wildlife. Grasslands are the most threatened habitat type in our region and are highly diverse overall. Secondly, staff is concerned about water rights associated with the constructed wetland. For example, if the created wetland intercepted groundwater, the subsequent land manager will need water rights adequate for augmenting the evaporative loss.~~

Staff recommends that the Board condition any approval on the following: that the applicant develop a grassland reclamation plan, consistent with the approved reclamation plan and reflective of the pre-mine vegetative potential of the site; the applicant shall submit a final reclamation plan for review by Boulder County Parks and Open Space staff at least six months prior to December 31, 2037; that the applicant shall make good faith reasonable efforts to incorporate into the final reclamation plan requests for modifications from Parks and Open Space staff.

Therefore, as conditioned, staff finds this criterion can be met.

- (3) ***Will be in accordance with the Boulder County Comprehensive Plan;***

There are multiple policies and goals in the Boulder County Comprehensive Plan which have been considered and evaluated in considering this criterion.

Element III: Cultural Resources Element

This Element of the Comprehensive Plan provides guidance on the County's efforts to identify, document, and protect the county's archaeological resources (either historic or prehistoric), historic buildings, sites, districts, and landscapes.

Policy CR 1.01 states: "*Boulder County shall continue researching and documenting the county's cultural resources including maintaining a comprehensive historic sites survey.*"

The original Special Use approval for the Dowe Flats quarry (SU-93-14) included a requirement that the operator produce and implement a cultural resources plan; this requirement was reemphasized and expanded during the site's 1-year compliance review by requiring the operator to catalog and document their cultural resource monitoring activities. As the requested extension for mining activities would occur in areas which have already been disturbed, no new cultural resource surveys are necessary. To ensure that those cultural resources already identified continue to be protected, staff recommend the Board include the following as a condition of approval: the 1994 Cultural Resource Management Plan shall remain active, for all cultural resources located on subject property, for the entirety of mining operations, including the annual monitoring required in the Cultural Resources Management Plan.

As conditioned, staff finds the proposal can be in accordance with this section of the Comprehensive Plan

Element V: Environmental Resources Element

This Element of the Comprehensive Plan provides guidance on the County's efforts to preserve, conserve and restore the unique and distinctive natural features, ecosystems and landscapes of the county using sound resource management principles and practices.

Policy ER 3.02 states: "*Boulder County shall encourage the removal of development rights from [Environmental Conservation Areas] through transfer, donation, acquisition, trade, or other incentives.*"

Policy ER 3.03 states: "*Development within ECAs shall be located and designed to minimize the cumulative impacts on the environmental resource values of ECAs.*"

Policy ER 3.05 states: "*Boulder County shall encourage and participate with the appropriate public entities and private landowners in the development of coordinated management plans to conserve, preserve and restore the environmental resource values of ECAs.*"

The Dowe Flats site is located in an area designated as the Rabbit Mountain Environmental Conservation Area as identified in the Comprehensive Plan. Boulder County currently holds options for the purchase of the land post-mining and post-reclamation, preventing future development on the site. The proposed extension keeps the County's option on the land and reduces the purchase price to zero dollars plus title and closing. The environmental resources of the site have already been impacted due to the mining activities on the site. Limiting mining activities to the area which has already been impacted, the cumulative impacts are minimized by preventing potential impacts to currently unaltered areas, whether in Boulder County or in other counties should the applicants pursue mining activities in surrounding areas. With the recommended conditions of approval under Criterion 2 above, specifically the condition to related to the grassland reclamation plan, the County will be able to help develop

a management plan that will preserve and restore the environmental resources of the Rabbit Mountain ECA more effectively than the restoration which would occur under the existing reclamation plan.

Policy ER 4.02 states: “Boulder County shall coordinate with local, state, and federal agencies and municipalities, as well as with willing private landowners, to protect natural resource values within Natural Landmarks and Natural Areas. This may include: identification of specific resources of concern including scenic values; recommendations for long-term management; mitigation of existing or foreseen impacts; or protection through acquisition of land interest.”

The northwest portion of the subject property is located within the Rabbit Mountain Natural Area as identified in the Comprehensive Plan (see Figure 3 below). The majority of the mining activities take place outside of the Natural Area, but some do occur at the very northern end of the property. The proposed extension of mining-activities will occur in these already impacted areas, minimizing the impact to the Natural Area as a whole. Additionally, the required reclamation of the subject property post-mining will significantly mitigate these impacts and will provide for the long-term management of the area.



Figure 3: The subject property with the Rabbit Mountain Natural Area shows in green.

With the County’s post-mining and post-reclamation purchase options on the subject property, and with the conditions of approval as recommended in Criterion 2 above, staff finds the proposal is in accordance with this section of the Comprehensive Plan.

Element VII: Geology Element

This Element of the Comprehensive Plan is intended to provide clear direction in the formulation and implementation of county land use decisions so far as geological factors are concerned. The goals and policies in this Element are also intended to: 1) move toward the attainment of the adopted goals for the Boulder County Comprehensive Plan as directed in

Goal Statements B.1 and B.2, and 2) fulfill the county's obligation to implement a Master Plan for Extraction of Commercial Mineral Deposits pursuant to 30-28-106(c), CRS.

Policy GE 2.05 states: "... the county shall prohibit or regulate, including by Special Use Review and the like, the open mining of any mineral or earthen material ..." Additionally, Policy GE 2.10 states: "In cooperation with the Colorado Mined Land Reclamation Board and its staff, the county shall require that all 'affected land' as defined by Colorado Statute, be reclaimed..."

The request by the applicant to amend the current approval for Open Mining at the Dowe Flats quarry site is being reviewed through the Special Use Review process. Both the existing approval and the requested extension amendment include post-mining reclamation of the affected land. As such, staff finds the proposal is in accordance with this section of the Comprehensive Plan.

Element X: Open Space Element

This Element of the Comprehensive Plan lays out the County's vision for understanding, preserving, and enjoying the county's natural heritage, and guidance on how to achieve these goals.

Policy OS 2.01 states: "Boulder County acquires real property rights to protect open space values and functions...." And Policy OS 2.02 states: "Boulder County acquires real estate interests in land, water, and minerals through appropriate real estate methods such as fee title, conservation easements, and trail easements."

As discussed previously, the County holds post-mining and post-reclamation purchase options on the subject property. ~~As part of the 15-year extension request, the applicant has committed to reducing the purchase price on these existing options, and to include new options on land owned by the applicant south of Highway 66 (see the submitted cover letter in Attachment A). Staff recommends that the following be included as a condition of any approval: the applicant shall sign a real estate option or other agreement ("New Agreement") that comprehensively documents the applicant's commitments in the application to grant Boulder County options to acquire land and easements from the applicant. The New Agreement shall include all water rights appurtenant to the option properties and reclamation requirements arising from this application, must be acceptable to the county in its sole discretion, and shall supersede and replace that certain Purchase Agreement, Lease and Option to Purchase dated July 11, 2002, recorded July 17, 2002, at Reception #2308598 in the office of the Clerk & Recorder of Boulder County. Additionally, the applicant shall work with Parks and Open Space staff to determine the appropriate seed mixture(s) for the required reclamation post-mining.~~

As conditioned, staff finds the proposal is in accordance with this section of the Comprehensive Plan.

Element XI: Public Health Element

The goals and policies in this Public Health element bolster and complement health-related content in other elements, and address additional topics related to public and environmental health.

Policy PH 1.01 states: "Boulder County recognizes the direct and secondary health impacts of outdoor air pollution produced by industrial, vehicular and other sources. The county collaborates with industry, state and neighboring governments to respond to and mitigate the

health impacts of poor air quality due to particulate matter, ground-level ozone, smoke from wildfires, greenhouse gases and other air pollutants.”

Many of the comments received from the public related to this docket express concerns about the health impacts of air pollution from the cement plant operations at the Lyons Quarry site. As discussed above, however, the Lyons Quarry and the cement plant operations are not part of this application or this review. As stated by the applicant, the plant would continue to be operated if the requested extension for the Dowe Flats quarry were to be rejected. ~~While the operations of the cement plant are not the subject of this review, the applicant has committed to the closure of the plant at the end of the requested 15-year extension for Dowe Flats. Having a defined and agreed upon timeframe for the closure of the plant, supports Policy PF 1.01. As such, staff finds the proposal is in accordance with this section of the Comprehensive Plan.~~

The extension as proposed and with recommended conditions is generally in accordance with the Boulder County Comprehensive plan. As discussed above, staff recommends several conditions of approval to ensure that the requested 15-year extension for the mining activities at Dowe Flats continue to be in accordance with specific polies of the Comprehensive Plan.

Therefore, as conditioned, staff finds this criterion can be met.

- (4) *Will not result in an over-intensive use of land or excessive depletion of natural resources. In evaluating the intensity of the use, the Board should consider the extent of the proposed development in relation to parcel size and the natural landscape/topography; the area of impermeable surface; the amount of blasting, grading or other alteration of the natural topography; the elimination or disruption of agricultural lands; the effect on significant natural areas and environmental resources; the disturbance of plant and animal habitat, and wildlife migration corridors; the relationship of the proposed development to natural hazards; and available mitigation measures such as the preservation of open lands, the addition or restoration of natural features and screening, the reduction or arrangement of structures and land disturbance, and the use of sustainable construction techniques, resource use, and transportation management.***

The location of the proposed extension of mining activities at the Dowe Flats quarry is a location which has already been impacted by past and current open mining activities as previously reviewed and approved. Many of the impacts addressed in the criterion have already occurred; these impacts include alterations to the natural landscape and topography, the effects on Rabbit Mountain Natural Area, and the disturbance of the plant and animal habitats. The proposed extension of these mining activities would not result in any increase in the size of the mining pit; rather, the mining activities would go deeper in the same location. While the proposed extension would not result in any expansion of the area of disturbance, it would, extend the period of the already existing impacts for an additional 15 years.

In considering these impacts and evaluating the proposed extension under this criterion, staff considered available mitigation measures, such as the preservation of open lands and the restoration of natural features. The County’s existing purchase options on the project area post-reclamation will help mitigate the impacts of the existing mining operations by ensuring that the land will be preserved as open space going forward. ~~The applicants’ commitment of record related to new options for County acquisition of lands south of Highway 66, after the closure of the Lyons Quarry and the cement plant and the required reclamation, will help mitigate the requested extension, as it will result in more open land being preserved.~~

A mitigation measure for the SU-93-14 approval and the Colorado State mining permit (M1993-041) require the mine operator to carry out reclamation of the land once mining activities have ceased. As addressed in the discussion of Criterion 2 above, the reclamation plans included in the application materials submitted are conceptual and are in need of refining to mitigate the impacts of the mining operations. Per the application materials (see Attachment A) and the applicant's subsequent reply to Boulder County Parks and Open Space's referral response (see Attachment C), the applicant agrees to work with Boulder County Parks and Open Space and the State of Colorado Division of Reclamation, Mining, and Safety to modify and refine the reclamation plans to ensure that the impacts resulting from the mining activities are appropriately mitigated post-mining operations at Dowe Flats.

Therefore, as conditioned in this Criterion and Criterion 2 above, staff finds this criterion can be met.

(5) *Will not have a material adverse effect on community capital improvement programs;*

There is no indication the proposal will have an adverse effect on community capital improvement programs, and no referral agency has responded with such a concern.

Therefore, staff finds this criterion is met.

(6) *Will not require a level of community facilities and services greater than that which is available;*

Staff does not anticipate the proposal will have an adverse effect on community facilities and services, and no agencies submitted a response voicing any concerns about community facilities or services.

Therefore, staff finds this criterion is met.

(7) *Will support a multimodal transportation system and not result in significant negative impacts to the transportation system or traffic hazards;*

There are two points of access to the subject property. The first is via State Highway 66 (SH 66), also known as Ute Highway, a Colorado Department of Transportation (CDOT) owned and maintained right-of-way (ROW). The second is via N. 53rd Street, a Boulder County owned and maintained ROW with the Functional Classification of Local. Legal access is demonstrated via adjacency to these public ROWs.

Boulder County Community Planning and Permitting staff with the Access and Engineering team reviewed the application material and concurs with the trip generation estimates provided in the Pre-Application Methodology Statements (PAMS) dated October 17, 2021 (see Attachment A) of approximately 20 Average Daily Trips (ADT) being generated by the continued mining and reclamation use on the subject property. Based on this information, staff does not anticipate the mining and reclamation use will have negative impacts to the surrounding transportation network.

Therefore, staff finds this criterion is met.

(8) *Will not cause significant air, odor, water, or noise pollution;*

The requested 15-year extension of the open mining activities are regulated, as noted below, to control and mitigate impacts on air, odor, water, and noise pollution. With the required

regulatory mitigation measures, the proposal extension will not have significant impacts. The existing mining activities operate under an Air Pollution Emissions Notice (APEN) and Storm Water Permit and associated stormwater management plan as issued by the State of Colorado Department of Public Health and Environment (CDPHE). There are no records of violations of these plans and permits at the Dowe Flats quarry site. To ensure that these potential impacts are tracked and monitored, staff recommends that the Board include the following as conditions of approval: the permit shall continue to be subject to the period review and assessment every five (5) years as required in the original approval of docket SU-93-14; the applicant shall ensure all site monitoring reports (air, water, wildlife, noise, et cetera) required by Boulder County and the State of Colorado Division of Mining, Reclamation and Safety are submitted to Boulder County Parks and Open Space.

The cement plant at the Lyons Quarry site has a significant impact on air quality and noise levels in the area. As discussed above, the current operations at the Lyons Quarry and the cement plant are not part of this application or this review. ~~As part of the proposal, however, the applicant has committed to close the cement plant at the end of the requested 15-year extension. Although staff does not have the ability to unilaterally stop the cement plant as part of this application, the applicant has agreed to do so as a commitment of record. If the requested extension is not approved, the applicant has stated that it intends to continue to operate the cement plant indefinitely, with no identified timeframe for its closure. While members of the public have questioned whether it would be financially feasible for the applicant to continue to operate the plant without access to raw materials from Dowe Flats, that analysis is beyond the purview of this application and the financial feasibility of importing materials is a decision that applicant would have to make. An increase in imported materials would certainly result in an increase in air, odor, and noise pollution due to an increase in truck traffic needed to bring those raw materials in from one or more alternate source locations.~~

~~Staff finds the continuation of the existing mining operations at Dowe Flats, and the closure and deconstruction of the cement plant at the end of the 15-year extension, would result in significantly less air, odor, water, and/or noise pollution than the indefinite operation of the cement plant at the Lyons Quarry site with the importation of raw materials. Staff recommends the Board include the following as a condition of approval: upon the expiration of the requested 15-year extension, as proposed by the applicant, the operator shall cease all operations at the Lyons Quarry site and carry out all required reclamation. This shall include the permanent closure and construction of the Lyons Quarry cement plant and the reclamation of the plant site.~~

Staff also finds that the air, odor, water, or noise pollutions impacts of the 15-years of continued mining at Dowe Flats are sufficiently mitigated by the state and federal environmental impact permitting requirements as discussed above. The noise impacts of the mining activities at Dowe Flats are mitigated by the earthen berms constructed around the site; the fact that mining activities occur deep within the existing mining pits and that the processing equipment on-site is enclosed.

Therefore, as conditioned, staff finds this criterion can be met.

(9) *Will be adequately buffered or screened to mitigate any undue visual impacts of the use;*

As noted above, the existing mining pit and most of the quarry operations are not visible from most public roads in the area due to the contours of the terrain and the distance of the mining operations from public roads. The quarry can be seen from Dakota Ridge Road, north of the Dowe Flats quarry location and from the hiking trails at the Rabbit Mountain Open Space.

Due to depth of the mining pit and distance from the mining pit to these public locations, however, most of the actual mining activities will not be visible to members of the public. Additionally, the area of site disturbance will not be increasing; instead, the excavation will be going deeper into the current pit.

Therefore, staff finds this criterion is met.

(10) *Will not otherwise be detrimental to the health, safety, or welfare of the present or future inhabitants of Boulder County;*

The requested 15-year extension of mining activities at the Dowe Flats quarry site would result in impacts to the health, safety, or welfare of the present or future inhabitants of Boulder County due to the inherent risks and environmental impacts of open mining. ~~These impacts to health, safety, and welfare can be sufficiently mitigated, however, by the recommended conditions of approval. Specifically, the closure of the cement plant at the end of the 15-year extension would remove a significant source of pollution, carbon-dioxide emissions, and heavy mining-related truck traffic on Highway 66.~~

~~In addition, the opportunity for the County to acquire a significant amount of land (post-reclamation), both north and south of Highway 66, as included in the applicants' commitment of record, would be a significant mitigation measure. It will provide the inhabitants of Boulder County with access to a significant amount of new open space for recreation.~~

Therefore, as conditioned, staff finds this criterion can be met.

(11) *Will establish an appropriate balance between current and future economic, environmental, and societal needs by minimizing the consumption and inefficient use of energy, materials, minerals, water, land, and other finite resources;*

With the recommended conditions of approval, allowing the continued Open Mining use at the Dowe Flats quarry will establish an appropriate balance between accommodating societal and economic needs for cement and concrete by providing for efficient extraction and transportation of raw materials, while mitigating the environmental impacts. ~~The environmental impacts are mitigated by the commitments of record and conditions of approval, specifically, options to purchase approximately 1,030 additional acres at or near the property, dedication of a permanent non-exclusive recreational trail easement along the south bank of the St. Vrain river, and the permanent closure of all operations at the Lyons quarry and cement plant at the end of the 15-year period.~~

Therefore, as conditioned, staff finds this criterion can be met.

(12) *Will not result in unreasonable risk of harm to people or property – both onsite and in the surrounding area – from natural hazards. Development or activity associated with the use must avoid natural hazards, including those on the subject property and those originating off-site with a reasonable likelihood of affecting the subject property. Natural hazards include, without limitation, expansive soils or claystone, subsiding soils, soil creep areas, or questionable soils where the safe-sustaining power of the soils is in doubt; landslides, mudslides, mudfalls, debris fans, unstable slopes, and rockfalls; flash flooding corridors, alluvial fans, floodways, floodplains, and flood-prone areas; and avalanche corridors; all as identified in the Comprehensive Plan Geologic Hazard and Constraint Areas Map or through the Special Review or Limited Impact Special Review process using the best available information. Best available information includes, without limitation, updated*

topographic or geologic data, Colorado Geologic Survey landslide or earth/debris flow data, interim floodplain mapping data, and creek planning studies.

The Boulder County Comprehensive Plan identifies a number of geologic hazards on the subject parcel including steeply dipping and heaving bedrock, debris flow susceptibility areas, rockfall susceptibility areas, landslide high susceptibility areas, and high swelling soil potential areas. However, all of these are directly associated with, or the result of, the Open Mining activities on the site. The area is not open to the public and access to the subject property is restricted. The extension of the mining activities would not result in any increased risk over existing conditions.

Additionally, the reclamation required under the Colorado State mining permit, SU-93-14, and the recommended conditions of approval regarding reclamation will significantly reduce these geological hazards once those reclamation requirements are completed.

Therefore, as conditioned, staff finds this criterion is met.

- (13) ***The proposed use shall not alter historic drainage patterns and/or flow rates unless the associated development includes acceptable mitigation measures to compensate for anticipated drainage impacts. The best available information should be used to evaluate these impacts, including without limitation the Boulder County Storm Drainage Criteria Manual, hydrologic evaluations to determine peak flows, floodplain mapping studies, updated topographic data, Colorado Geologic Survey landslide, earth/debris flow data, and creek planning studies, all as applicable given the context of the subject property and the application.***

The original, historic drainage patterns on the subject property have already been significantly impacted. The requested extension of mining activities will not result in any additional changes to drainage patterns in the long-run.

The required reclamation of the area post-mining will significantly mitigate these impacts once mining activities have ceased. ~~Boulder County Parks and Open Space staff have reviewed the proposed reclamation plan and have determined that they do not support the creation of wetlands and open water features in the reclamation plan. Specifically, staff finds that returning the site as much as possible to its pre-mine landscape as grassland is the best for long-term ecosystem sustainability.~~ With the recommended conditions included under criteria 2, 3, and 12 above, the impacts to historic drainage patterns will be acceptably mitigated.

Therefore, as conditioned, staff finds this criterion is met.

RECOMMENDATION

Staff has determined that the proposal can meet all the applicable criteria of the Boulder County Land Use Code for Special Review. Therefore, staff recommends that the Planning Commission recommend that the Board of County Commissioners ***conditionally approve docket SU-22-0003 Dowe Flats Mining and Reclamation Extension*** with the following conditions:

1. The applicants shall provide a Development Agreement, for review and approval by County staff, prior to the issuance of a license or permits by the Boulder County Community Planning & Permitting Department and prior to the recordation of said agreement within one year of approval.

2. The 1994 Cultural Resource Management Plan shall remain active, for all cultural resources located on the subject property, for the entirety of mining operations, including the annual monitoring required in the Cultural Resources Management Plan.
3. The applicant shall obtain any new, or amend any existing, state, and federal permits as appropriate and necessary, and maintain any and all such permits for the duration of mining activities at the subject property.
4. The applicant shall develop a grassland reclamation plan, consistent with the approved reclamation plan and reflective of the pre-mine vegetative potential of the site.
5. The applicant shall submit a final reclamation plan for review by Boulder County Parks and Open Space staff at least six months prior to December 31, 2037.
6. The applicant shall make good faith reasonable efforts to incorporate into the final reclamation plan requests for modifications from Boulder County Parks and Open Space staff.
7. ~~The applicant shall sign a real estate option or other agreement ("New Agreement") that comprehensively documents the applicant's commitments in the application to grant Boulder County options to acquire land and easements from the applicant. The New Agreement shall be acceptable to the county in its sole discretion.~~
8. The applicant shall work with Boulder County Parks and Open Space staff to determine the appropriate seed mixture(s) for the required post-mining reclamation.
9. The permit shall continue to be subject to the periodic review and assessment every five (5) years as required in the original approval of docket SU-93-14.
10. The applicant shall ensure all site monitoring reports (air, water, wildlife, noise, et cetera) required by Boulder County and the State of Colorado Division of Mining, Reclamation and Safety are submitted to Boulder County Parks and Open Space.
11. ~~At the expiration of the 15-year extension, as dated from the date of approval by the Board of County Commissioners, the operator shall cease all operations at the Lyons Quarry site and carry out all required reclamation. This shall include the permanent closure and deconstruction of the Lyons Quarry cement plant and the reclamation of the plant site.~~
12. The applicants shall be subject to the terms, conditions, and commitments of record and in the file for docket **SU-22-0003: Dowe Flats Mining and Reclamation Extension**.